Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering and Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: Linear Regression**	
Experiment No:- 2	**Date:-**	**Enrollment No:-**92301733001

1. Simple Linear Regression

Code:-

```
import numpy as np import
matplotlib.pyplot as plt class
SimpleLinearRegression:     def
__init__(self):
        self.coefficients_ = None
self.intercept_ = None
self.r2score_ = None
def fit(self, X, y):
        n = len(X)
        X_b = np.c_[np.ones((n,1)), X]        self.coefficients_ =
np.linalg.inv(X_b.T.dot(X_b)).dot(X_b.T).dot(y)        self.intercept_ =
self.coefficients_[0]        y_pred = X_b.dot(self.coefficients_)
        # R² Score        self.r2score_ = 1 - (np.sum((y - y_pred)**2) / np.sum((y -
np.mean(y))**2))        self.y_pred_ = y_pred            def predict(self, X):
        X_b = np.c_[np.ones((len(X),1)), X]
return X_b.dot(self.coefficients_)
        # Dataset
X_simple = np.array([1,2,3,4,5,6,7,8,9,10]).reshape(-1,1) y_simple
= np.array([2,4,5,4,5,7,8,9,10,12])
# Fit Model slr =
SimpleLinearRegression()
slr.fit(X_simple, y_simple)
# Print metrics print(f"Simple LR Coefficients:
{slr.coefficients_}") print(f"R² Score:
{slr.r2score_:.2f}")
```

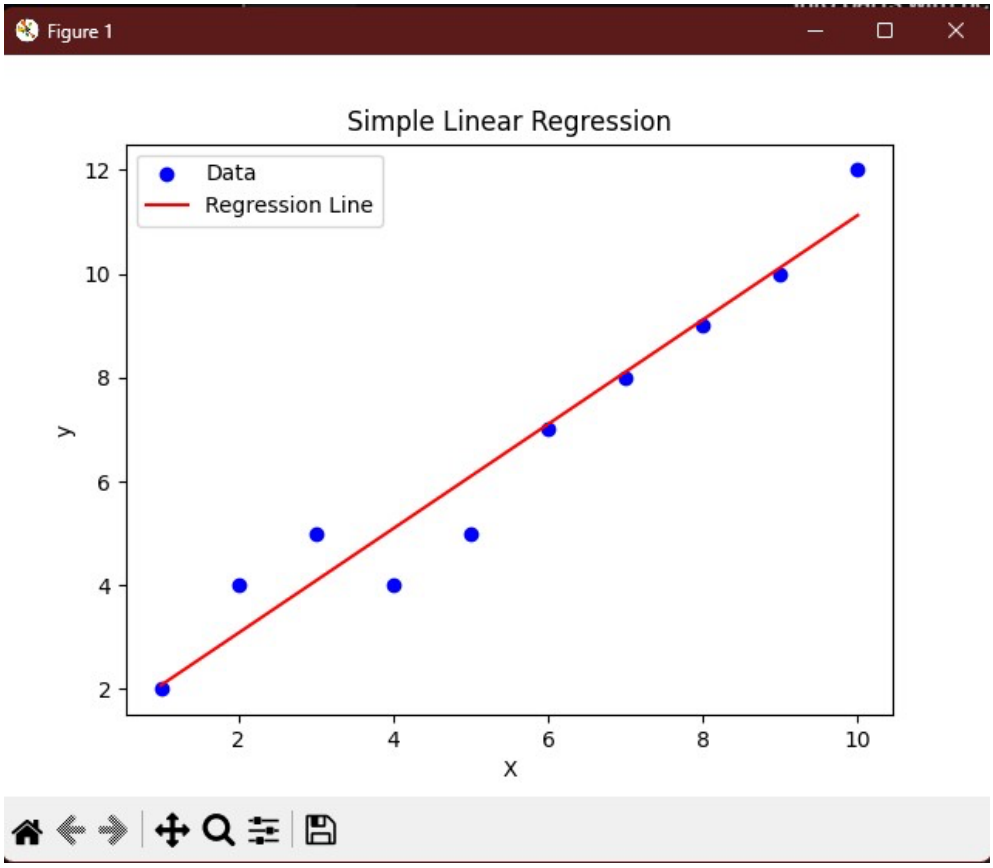
Visualize plt.scatter(X_simple, y_simple, color='blue',

label='Data') plt.plot(X_simple, slr.y_pred_, color='red',

label='Regression Line') plt.title("Simple Linear Regression")

plt.xlabel("X") plt.ylabel("y") plt.legend() plt.show()

Output:-



2. Multiple Linear Regression

Code:- import numpy as np import

matplotlib.pyplot as plt from

mpl_toolkits.mplot3d import Axes3D class

MultipleLinearRegression: def

__init__(self):

```
    self.coefficients_ = None
self.intercept_ = None
self.r2score_ = None
def fit(self, X, y):        n =
X.shape[0]
    X_b = np.c_[np.ones((n,1)), X]        self.coefficients_ =
np.linalg.inv(X_b.T.dot(X_b)).dot(X_b.T).dot(y)        self.intercept_ =
self.coefficients_[0]        y_pred = X_b.dot(self.coefficients_)
self.r2score_ = 1 - (np.sum((y - y_pred)**2)/np.sum((y - np.mean(y))**2))
self.y_pred_ = y_pred            def predict(self, X):
    X_b = np.c_[np.ones((X.shape[0],1)), X]
return X_b.dot(self.coefficients_)
np.random.seed(0)
X1 = np.random.randint(1, 11, 15)
X2 = np.random.randint(1, 11, 15) X_multi =
np.column_stack((X1, X2)) y_multi = 1 + 2*X1 +
3*X2 + np.random.randn(15)*2 mlr =
MultipleLinearRegression() mlr.fit(X_multi, y_multi)
print(f"Multiple LR Coefficients:
{mlr.coefficients_}") print(f"R² Score:
{mlr.r2score_:.2f}") fig = plt.figure(figsize=(10,7)) ax
= fig.add_subplot(111, projection='3d')
ax.scatter(X_multi[:,0], X_multi[:,1], y_multi,
color='blue', label='Data') x1_surf, x2_surf =
np.meshgrid(    np.linspace(X_multi[:,0].min(),
X_multi[:,0].max(), 10),
```

```
np.linspace(X_multi[:,1].min(), X_multi[:,1].max(),
10)
)
pred_surf = mlr.predict(np.c_[x1_surf.ravel(), x2_surf.ravel()]).reshape(x1_surf.shape)
ax.plot_surface(x1_surf, x2_surf, pred_surf, color='red', alpha=0.5, rstride=1, cstride=1)
ax.set_xlabel('X1') ax.set_ylabel('X2') ax.set_zlabel('y') ax.set_title("Multiple Linear
Regression with Regression Plane") ax.legend() plt.show()
```

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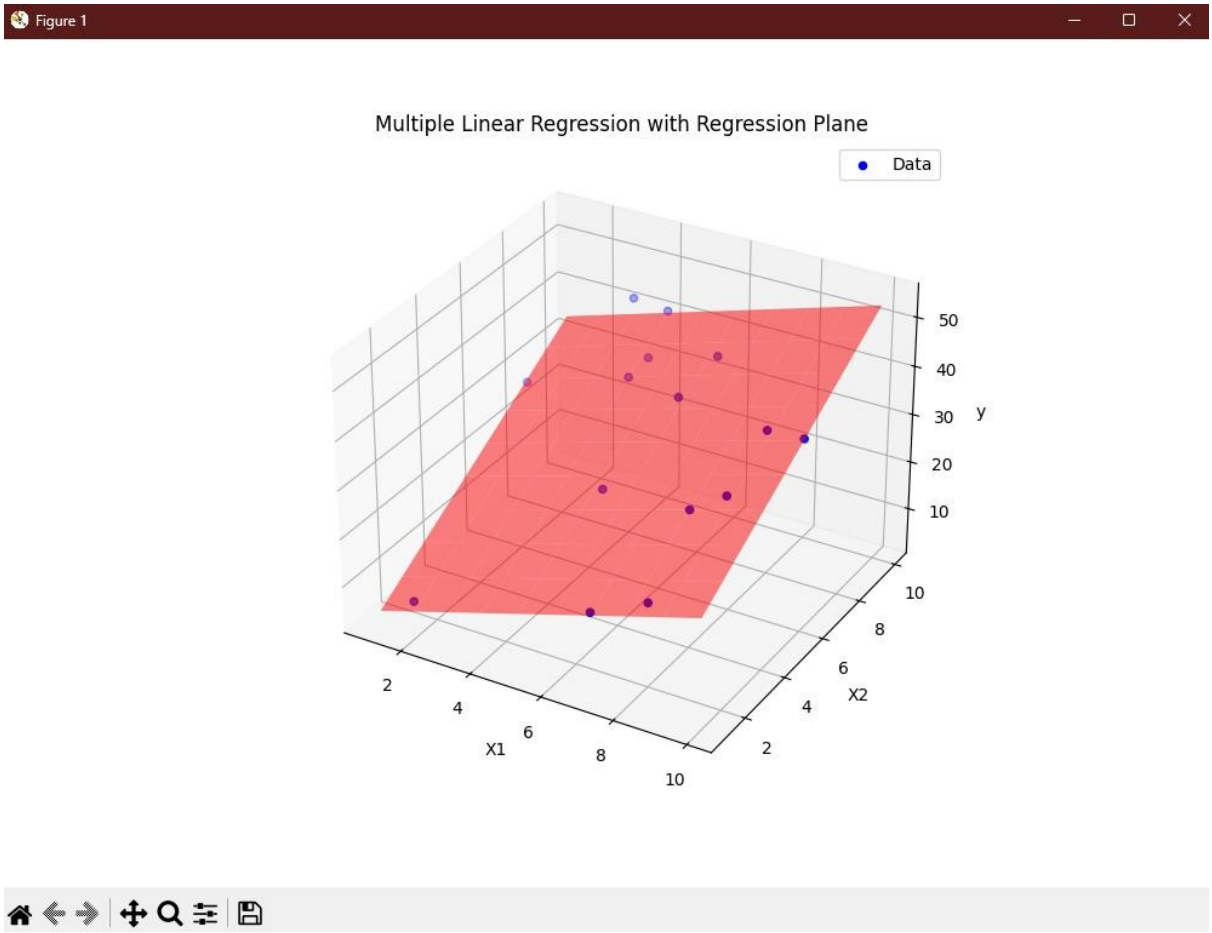
Output:-



3. Polynomial Regression

Code:- import numpy as np import matplotlib.pyplot

as plt from sklearn.preprocessing import

PolynomialFeatures class PolynomialRegression:

def __init__(self, degree=2): self.degree = degree

self.coefficients_ = None self.intercept_ = None

self.r2score_ = None self.poly_features = None

def fit(self, X, y):

self.poly_features = PolynomialFeatures(degree=self.degree, include_bias=True)

X_poly = self.poly_features.fit_transform(X) self.coefficients_ =

np.linalg.inv(X_poly.T.dot(X_poly)).dot(X_poly.T).dot(y) self.intercept_ =

self.coefficients_[0] y_pred = X_poly.dot(self.coefficients_) self.r2score_ =

1 - (np.sum((y - y_pred)**2) / np.sum((y - np.mean(y))**2)) self.y_pred_ =

y_pred def predict(self, X):

X_poly = self.poly_features.transform(X)

return X_poly.dot(self.coefficients_)

np.random.seed(42)

X_poly = np.random.rand(50,1)*6 - 3 y_poly = 0.5*X_poly**2 +

X_poly + 2 + np.random.randn(50,1)*0.5 y_poly = y_poly.flatten()

pr = PolynomialRegression(degree=2) pr.fit(X_poly, y_poly)

print(f"Polynomial LR Coefficients: {pr.coefficients_}") print(f"R²

Score: {pr.r2score_:.2f}") plt.scatter(X_poly, y_poly, color='blue',

label='Data')

plt.plot(np.sort(X_poly, axis=0), pr.y_pred_[np.argsort(X_poly, axis=0)], color='red',

label='Polynomial Fit') plt.title("Polynomial Regression") plt.xlabel("X")

plt.ylabel("y") plt.legend() plt.show()

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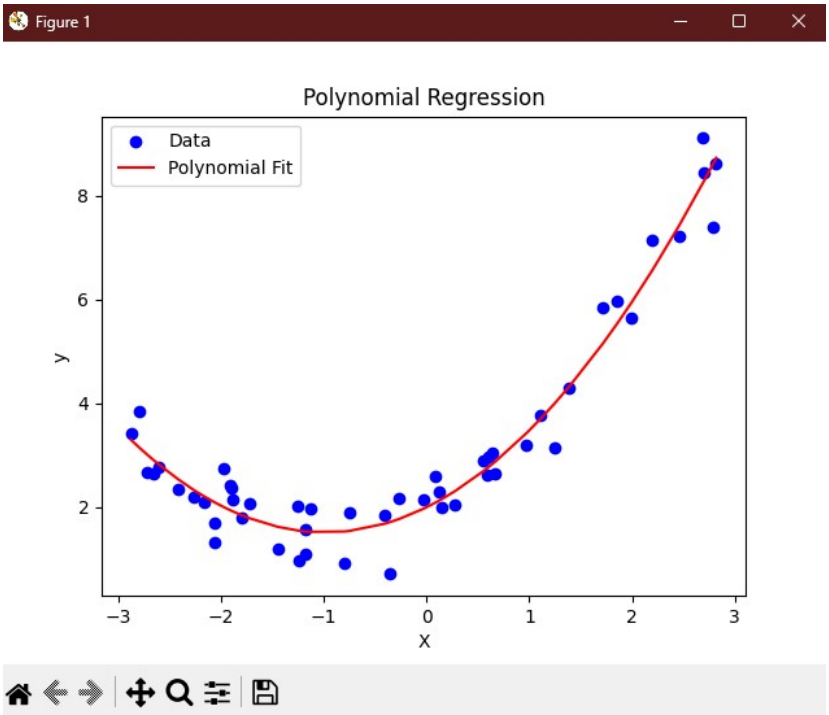
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